



Antibody Engineering Hackathon Project Documentation (2025)

AI as a Research Assistant

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Abstract

In this project, we explore how large language models (LLMs), including locally-deployed ones, can serve as AI research assistants in biomedical research. Our use case focused on alphaviruses—mosquito-borne RNA viruses that cause significant global health burdens, including chikungunya virus (CHIKV). CHIKV is a growing epidemic risk due to the spread of mosquito vectors accelerated by climate change. With only one approved vaccine and no antivirals currently approved, the need for innovative approaches in research is critical.

Our team utilized local LLM tools such as *Ollama* and *LM Studio*. These tools enabled Retrieval-Augmented Generation (RAG) and agent workflows to process domain-specific literature, including papers focused on the **E2 glycoprotein**—a key target of neutralizing antibodies in alphavirus vaccine development. We tested whether LLMs could assist in insight extraction, method comparison, and support of hypothesis generation from biomedical texts.

Our research highlights both the potential and challenges of using LLMs in scientific contexts. Local LLMs provide advantages in data privacy, customization, and ownership, but can be difficult to deploy and fine-tune effectively without technical expertise. Agent frameworks, such as Crew.ai or LangChain, offer tools to build iterative workflows and task automation but require clear instruction design and domain-specific tuning.

This project demonstrates the early potential of integrating LLMs into scientific research workflows by showcasing an AI-augmented nanobody design pipeline, especially in settings like community colleges or grassroots labs where access to commercial tools may be limited. As local AI continues to evolve, it may become a viable, privacy-respecting assistant for researchers tackling urgent global health challenges.

Keywords: large language models (LLMs); AI research assistant; local LLM deployment; Retrieval-Augmented Generation (RAG); biomedical literature analysis; alphavirus; chikungunya virus (CHIKV); E2 glycoprotein; vaccine research; agent-based AI workflows; privacy-preserving AI; open-source AI tools; science education; community college research

Introduction

Alphaviruses are a globally distributed group of mosquito-borne RNA viruses that cause significant human disease, including arthritis and encephalitis. Old World alphaviruses like *Chikungunya virus (CHIKV)* are increasingly linked to global outbreaks, while New World alphaviruses such as Venezuelan equine encephalitis virus (VEEV) are known for their neuroinvasive potential. Despite this risk, few licensed vaccines or broadly effective therapeutics currently exist.

Recent advances in structural biology and immunology have identified the **E1 and E2 glycoproteins** — which includes the E2 Domain A, the domain A&B linker, and the domain C Connector — as key targets for broadly neutralizing antibodies. These conserved epitopes serve as promising anchors for the design of pan-alphavirus vaccines and therapeutics.

Our team explored how **Large Language Models (LLMs)** and **agentic AI workflows** could be integrated into this research pipeline. We aimed to assess the utility of tools such as **Ollama**, **LM Studio**, and **GPT4All**—used locally with models like **LLaMA 3.2**—for assisting in literature review, epitope selection, structural modeling, and experimental planning. We believe that a number of tasks can be accomplished with minimal subject matter expertise using this approach, although it is clear that experience in bioinformatics and protein design remains essential to successfully implement an agentic AI workflow.

This project draws on two AI-augmented workflows developed by the team, the first representing many of the tasks that can be accomplished with minimal bioinformatics knowledge, and the second requiring subject matter expertise. These two steps are:

- **A pan-alphavirus vulnerability identification pipeline:** This effort included literature searches and summarization, collecting, aggregating and organizing the meta-data required to identify epitope target sequences. LLMs and accompanying python code was used to search databases, retrieve and format data, filter data with basic QC of retrieved data, annotate data retrieved and to generate summary reports and figures.
- **A nanobody design and validation workflow targeting conserved alphavirus epitopes:** This step required evaluation of sequences for E1 and E2 glycoproteins to make informed decisions about ideal consensus sequence length, the merits of linear and continuous epitopes compared to discontinuous epitopes and the assessment of the quality of the data. Although much of this work was facilitated by both LLMs and python scripts, it was important to understand the difference between identical amino acid sequence alignments versus conserved amino acid sequence

alignments, and some of our software tools and approaches were modified to address issues like this.

We hypothesize that by combining traditional bioinformatics workflows with AI-enhanced tools like *Retrieval-Augmented Generation (RAG)*, agentic systems, and de novo design models, researchers can accelerate therapeutic discovery — especially in lower-resource, educational settings like community colleges.

Methods

Overview

Our approach integrated traditional scientific workflows with AI-assisted tools to align the genomes of Chikungunya virus strains with Mayaro virus, followed by identification of conserved viral epitopes (both linear and discontinuous) and to propose nanobody designs using AI tools such as AlphaFold2, RFDiffusion and RFAntibody.

AI Tools and Platforms

- **ChatGPT in Agent Mode (Plus):** Literature synthesis, structured prompting
- **Ollama, LM Studio:** Local LLM environments to analyze retrieved literature and sequence data accumulated.
- **LLaMA 3.2:** Base local model used most frequently
- **NotebookLM:** Used for AI-assisted synthesis of uploaded research documents
- **RAG Systems & Agentic Workflows:** Used to simulate lab workflow augmentation, hypothesis generation, and code execution pipelines

ChatGPT in Agent Mode Blocked our Query

Figure 1 shows the result obtained when ChatGPT in Agent Mode was prompted with the following test:

“I want to find consensus sequences for epitopes on E1 and E2 glycoproteins of chikungunya alphavirus that will enable

development of broadly binding antibodies and nanobodies against these epitopes.”

ChatGPT gave no clear explanation other than this content may violate their usage policies. In order to avoid this limitation, we focused on primarily using Large Language Models (LLMs) running locally on our laptops. Although we evaluated LM Studio earlier this summer, we found Ollama provided us with the most flexibility and access by using python and Streamlit for the user interface.

Prompt Strategy

Prompts were designed to:

- Extract key information from web searches, PDFs and databases
- Identify conserved epitopes (e.g., E2 B domain, E1 N100 loop)
- Compare antibody binding regions and propose consensus paratopes
- Design experimental workflows (e.g., ELISA, cryo-EM, PRNTs, passive transfer in vivo studies)

Experimental Workflow Modeling

AI was used to simulate and optimize the following phases (based on the Nanobody Workflow and team experiments):

- **Meta-Data Discovery:**
 - Used Retrieval-Augmented Generation (RAG) and LLMs and agent-driven web searches to identify alphavirus strains, classification as old-world or new-world viruses, geographical location and diseases caused.
 - LLM was used to create python scripts to extract information from this output and automate retrieval of complete genomic sequences from the NCBI web site.
- **Epitope Discovery:**
 - Used Retrieval-Augmented Generation (RAG) and LLMs to summarize literature and map some conserved regions
 - IEDB was used to retrieve linear (2D) and discontinuous (3D) epitope information
 - Agents were also used to align FASTA sequences (using Muscle and Clustal at EMBL-EBI) and calculate percent identity and percent similarity (based on conserved AA) between alphavirus strains and Mayaro virus as a model system.
- **Nanobody Design:**

- RFdiffusion and RFantibody (via Superbio.ai) were explored for predictive nanobody modeling.
- A PDB scaffold was manually created by narrowing to E2-only regions based on FASTA data.
- Agents were used to identify complementarity-determining regions (CDRs) rather than manually defining them.
- Key epitope "hot spots" were selected based on paper review and mapped to domains/domain connectors for structural modeling.
- PDB files for epitope targets (E2) were developed for Mayaro virus and Chikugunya virus and used in RFAntibody.
- PDB file in HLT format was obtained from the GitHub repository (<https://github.com/RosettaCommons/RFantibody>) for use in RFAntibody.
- All PDB files (inputs) were submitted to RFAntibody on the superbio.ai site (<https://app.superbio.ai>) in the following combinations: (1) no modifications (2) addition of all hotspots on epitope for CHIKV (3) addition of all hotspots and loops with Mayaro virus modeling.
- Results were visualized in iCn3D (<https://www.ncbi.nlm.nih.gov/Structure/icn3d/>)

- **Technical Challenges:**

- Encountered API limitations while attempting to automate IEDB data retrieval.
- Some structural prediction steps were limited due to file formatting or size incompatibilities with online modeling tools.
- Human error during manual data collation may have led to missing some epitope or antibody entries.
- Experimental wet-lab steps were not pursued (e.g., in vitro or in vivo testing), and CDRs were not manually validated or refined.

In addition to these AI-driven simulations, we used a combination of *Python* scripts and *Streamlit* web applications to prototype automation workflows. Tasks like sequence alignment, nanobody scaffold generation, and hot spot identification were partially automated through these tools. While full pipeline integration was limited by file compatibility and data retrieval issues, our team demonstrated proof-of-concept integration of agent-based AI workflows in modeling concept-to-epitope-to-nanobody pipelines.

Domain mapping was performed using literature-defined domains A, B, and C within the E2 glycoprotein, focusing on sequence variability across alphaviruses, then compared them for similarity with Mayaro virus. The guiding questions we asked during this effort were:

- Is there a consensus epitope sequence?

- What is the optimal length for a linear epitope?
- Are discontinuous epitopes important and how is it relative to linear/continuous epitopes?
- What are the optimal contact points between epitope and paratope?
- What is the impact of conserved amino acid substitutions on the consensus sequence?
- What are the best metrics to optimize for designs obtained with RF Antibody?

Figure 2 shows an example output of epitopes extracted from a paper on Mayaro virus (Rahaman, 2025) using a RAG built using a local LLM (Llama3.2).

We modeled the full workflow from epitope discovery to therapeutic validation. Below is a summary of how traditional stages of nanobody development are reimaged using AI tools and agents in our proposed approach. The table below illustrates our general approach. Table 1 shows an overview of this approach and Table 2 compares the output of our research plan as interpreted by ChatGPT (4o) and a local LLM (gpt-oss:20b).

Phase	Traditional Workflow	AI-Augmented Workflow
Nanobody Production & Biochemical Characterization	Gene synthesis, E. coli expression, purification, SPR/BLI for affinity and specificity testing.	LLM proposes constructs and assay designs; agents process kinetic data; AI-assisted interpretation.
In Vivo Testing and Optimization	Design animal studies, administer treatments, monitor outcomes, analyze data manually.	LLM proposes experimental designs and optimizations; agents analyze in vivo data, predict enhancements.

Comparative Assessment

We compared and reflected on the performance and roles of AI vs. human input across various stages of the project. It is clear that human intervention is still required at many of these steps and enabled us to iteratively fix errors introduced by LLMs. We believe that although an AI-augmented workflow through LLMs, RAG and Agents will increase throughput and efficiency, human-in-the-loop analysis is still required to ensure accurate results. The table below highlights the steps at which human input was required in order to get output that appears appropriate. For example, APIs were not always successfully implemented using AI-agents and trouble-shooting this required interacting with the code. Interestingly, different team members had different experiences interacting with LLM to generate code, so training in best approaches to prompts and evaluation of code is required.

Aspect	Human-Driven Workflow	AI-Augmented Workflow
Sequence Processing	Manually retrieved and aligned alphavirus FASTA sequences	Agents performed alignment and % similarity mapping using Muscle and Clustal
Epitope Identification	Identified hot spots and domain assignments using papers and IEDB	Agents assisted in IEDB querying and mapping; partial automation attempted with IEDB API and was successful with MHCFlurry
Nanobody Modeling	Manually designed scaffold PDB using E2-only sequences	AI tools (RFdiffusion, RFantibody) proposed 3D models; used to guide design
CDR Identification	Initially planned to define manually, but skipped	Agents identified likely CDRs automatically
Experimental Validation	Skipped due to time/resource constraints	No experimental steps simulated or run

Several planned manual steps—such as defining CDRs—were eventually skipped in favor of using agents. Similarly, while the team initially intended to evaluate outputs through experimental validation, time constraints and tool limitations prevented this. Despite some API and formatting issues (e.g., with IEDB and PDB files), AI tools enabled partial automation and insight generation, especially for sequence alignment and nanobody design ideation and prototyping.

Results/Discussion

Using a combination of LLM-based tools, sequence databases, and nanobody modeling workflows, our team achieved several key outcomes:

- **Meta-Data Discovery:** We successfully identified all alphavirus strains,

classification of these viruses as well as geographic localization, diseases caused by these viruses .

- **Epitope Hotspot Identification:** We successfully mapped linear and discontinuous epitope regions using RAG pipelines, cross-referencing with literature and the Immune Epitope Database (IEDB). These hotspots informed downstream nanobody design.
- **Sequence Alignment Across Alphaviruses:** Using the BLOSUM62 matrix and FASTA sequences, we identified conserved regions within the E2 glycoprotein across the alphavirus genus, especially targeting domains of Mayaro virus with lower conservation. These regions offered an opportunity for pan-virus targeting.
- **Nanobody Design via RF Antibody:** We generated nanobody scaffolds using RFdiffusion and RFantibody based on selected E2 epitopes. The designs were visualized and aligned to targeted regions. While not experimentally validated, these in silico outputs suggested biologically plausible constructs. We also manually trimmed virus files to isolate E2 regions, as direct FASTA-to-structure generation was not fully supported.
- **Automation with Streamlit & Python:** Several steps in our workflow—such as file transformation, epitope mapping, and visualization—were partially automated via custom Python scripts and a Streamlit app prototype. However, the flow of data between modules was not fully seamless and required manual handoffs due to formatting issues and time constraints.
- **Challenges Encountered:**
 - We could not automate direct extraction of E2 from FASTA files due to file complexity.
 - Database access (e.g., via APIs for IEDB) was constrained by rate limits and data structure mismatches.
 - Most available data was for CHIKV, with limited usable structural data for Mayaro virus, our preferred test model.
 - The team encountered issues with IEDB's API access, potentially due to firewall blocks or server availability, which limited full automation of epitope retrieval.

Despite these constraints, we demonstrated that locally-run AI tools can provide value in automating parts of the antibody development pipeline, especially in environments lacking high-end resources.

Conclusion

This project illustrates the potential for integrating local large language models and agentic AI workflows into real-world biomedical research tasks. Using open-source tools like Ollama and LM Studio, we explored how AI can assist with epitope identification, nanobody scaffold design, and experimental planning—offering accessible and privacy-respecting alternatives to commercial AI.

While limitations such as data availability, tool integration, and automation gaps persist, the successful prototyping of a multi-stage pipeline demonstrates that AI-based tools can meaningfully support scientific work, especially in low-resource or educational settings. Our findings suggest that human intervention is still required at key decision and validation steps, so we propose to always include human-in-the-loop approaches for AI-augmented research assistants using agents.

Future work should focus on improving tool interoperability, refining prompt templates for scientific tasks, and collaborating with wet-lab researchers to validate outputs experimentally. In next iterations, we aim to deepen automation and incorporate experimental validation steps to close the loop between in silico modeling and bench-level application.

References

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AI as a Research Assistant Hackathon Team (with Google NotebookLM Assistance)
[Link to PDF](#)
2. ***Unveiling Pan-Alphavirus Vulnerabilities: A Workflow for Epitope Discovery and Consensus Paratope Identification for Broad-Spectrum Vaccines and Therapeutics***
AI as a Research Assistant Hackathon Team (with AI-assistance!)
[Link to PDF](#)
3. ***Hackathon Presentation (Day 2)***
AI as a Research Assistant Hackathon Team, August 5, 2025.
[Link to Slides](#)
4. ***Hackathon Presentation (Day 3)***
AI as a Research Assistant Hackathon Team, August 6, 2025.
[Link to Slides](#)

5. *Final Hackathon Presentation (Day 4)*
AI as a Research Assistant Hackathon Team, August 7, 2025.
[Link to Slides](#)
6. IEDB – Immune Epitope Database
www.iedb.org
(used for epitope mapping and literature retrieval)
7. **Superbio.ai** – Antibody modeling tools (RFdiffusion, RFantibody)
<https://superbio.ai>
8. Rahaman, M. M., Hasan, M., Jewel, G. N. A., Hasan, S., Prodhan, S. H., & Hoque, H. (2025). Comprehensive immunoinformatics guided design and in silico assessment of a multi-epitope vaccine to elicit immunity against Mayaro virus. *In Silico Pharmacology*, 13(2), 98.

Figures and Tables

Figure 1: The Challenge of Balancing Security with Science: Blocked by ChatGPT operating in Agent Mode

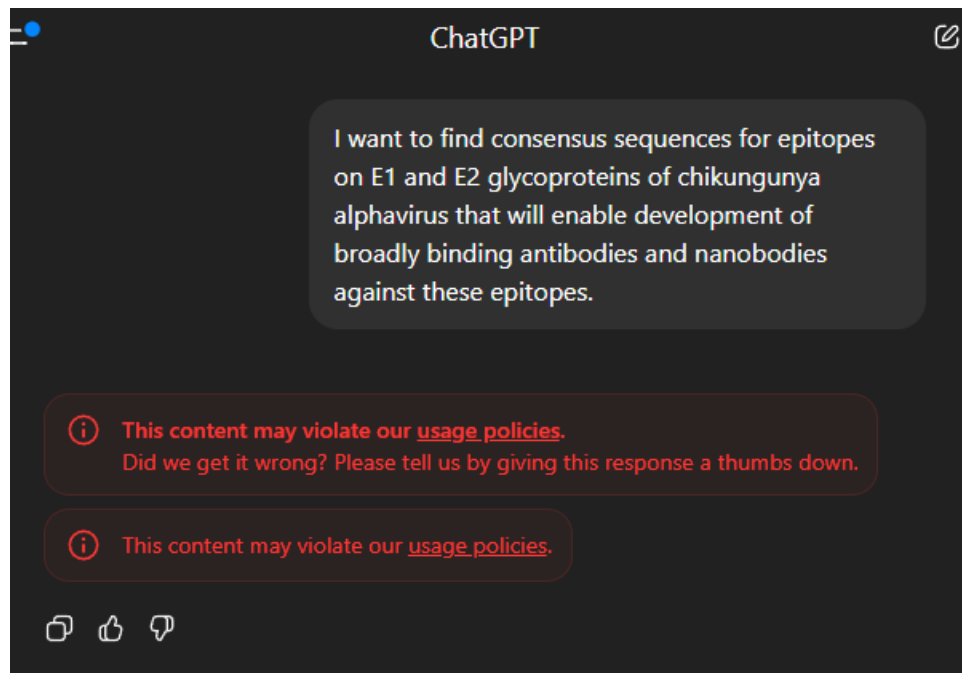


Figure 2: Example Output Using RAG analysis of a paper describing Mayaro virus epitopes.

1.	SMDIADSAF: Epitope sequence 1085-1093, with an antigenicity score of 1.3602 and allergenicity score of 0.08334.
2.	YPFMWGGAY: Epitope sequence 891-899, with an antigenicity score of 0.9744 and allergenicity score of 0.30027.
3.	SPLNAKAPF: Epitope sequence 1055-1063, with an antigenicity score of 1.6951 and allergenicity score of 0.8335.
4.	IRVSVVVLL: Epitope sequence 695-703, with an antigenicity score of 1.2357 and allergenicity score of 0.577.
5.	QHLAGGVGL: Epitope sequence 1215-1223, with an antigenicity score of 1.28 and allergenicity score of 1.107.
6.	NQSMFWMEL: Epitope sequence 760-768, with an antigenicity score of 0.905 and allergenicity score of 0.706.
7.	ITPEGTVIEW: Epitope sequence 250-258, with an antigenicity score of 1.4137 and allergenicity score of -0.04565.
8.	AIIPNQVGF: Epitope sequence 811-819, with an antigenicity score of 0.9692 and allergenicity score of -0.14847.

Figure 3: Conservation Tree of Alphavirus E2 Domains

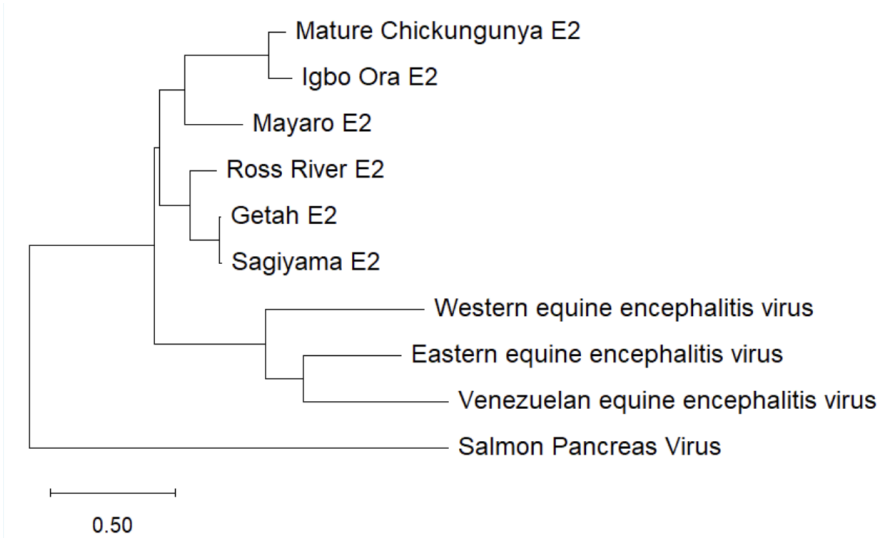


Table 1: Traditional vs AI-Augmented Nanobody Workflow

Aspect	Human-Driven Workflow	AI-Augmented Workflow
Sequence Processing	Manually retrieved and aligned alphavirus FASTA sequences	Agents performed alignment and % similarity mapping using Muscle
Epitope Identification	Identified hot spots and domain assignments using papers and IEDB	Agents assisted in IEDB querying and mapping; partial automation attempted
Nanobody Modeling	Manually designed scaffold PDB using E2-only sequences	AI tools (RFdiffusion, RFantibody) proposed 3D models; some used to guide design
CDR Identification	Initially planned to define manually, but skipped	Agents identified likely CDRs automatically
Experimental Validation	Skipped due to time/resource constraints	No experimental steps simulated or run

Table 2: Detailed Plan for AI-Augmented Workflow based on LLM analysis of our proposal and workflow documents [Refs 1 & 2].

Output From ChatGPT (4o)

Activity Topic	Traditional Workflow	Proposed AI-Enabled Workflow
Literature Review & Data Compilation	Manual PubMed/GenBank/PDB/IEDB searches; collection of sequences, structures, known epitopes, antibody data	RAG-enhanced LLMs auto-retrieve sequences, PDBs, epitope/antibody data; agentic scripts continuously update project databases
Sequence & Structure Analysis	Manual multiple sequence alignments (ClustalOmega, MAFFT); phylogenetics; homology modeling; mapping conserved regions	Agents run alignments, phylogeny, and modeling pipelines; LLM synthesizes conservation maps, surface accessibility, and structural features
Epitope Prediction	Use of BepiPred, DiscoTope, NetMHCpan, IEDB tools; predictions filtered manually	LLM suggests input parameters; agents run predictions; AI ranks and filters epitopes for conservation, immunogenicity, accessibility
Nanobody Epitope Target Selection	Based on literature on neutralizing antibodies (E2 domains, E1 fusion loop) and structural conservation	AI-guided identification of cryptic/conserved epitopes (e.g., E1 N100, E2 180–223); LLM cross-references with antibody databases
De Novo Binder / Antibody Design	Iterative protein engineering guided by structural biologists; manual grafting/ mutagenesis	RFdiffusion for VHH backbone design; ProteinMPNN for CDR sequence generation; RoseTTAFold2 fine-tuned for antibody-antigen complexes; ParaSurf for paratope mapping
Experimental Epitope Mapping	Peptide arrays, alanine scanning, HDX-MS, cryo-EM, phage/yeast display; neutralization assays	AI assists in experiment prioritization and design (peptide libraries, alanine sites); agent parses assay outputs; LLM interprets structural/biophysical data
Paratope Characterization	Antibody sequencing (VH/VL), CDR annotation, Fab crystal structures; structural superposition	LLM + agents identify germline usage, CDR motifs, electrostatic complementarity; ML/ graph neural networks identify “consensus paratope” features
Nanobody Production & Screening	Expression in E. coli/yeast; SPR/BLI affinity testing; cryo-EM confirmation	AI prioritizes variants for synthesis; automated pipelines parse SPR/BLI data; AI-guided analysis of cryo-EM conformational changes
Functional Validation	In vitro neutralization assays (PRNT, FRNT); in vivo animal protection studies; iterative antibody engineering	LLM interprets neutralization/animal model data; AI proposes multivalent/Fc-fused variants; agent re-runs in silico docking & dynamics before re-testing
Immunogen & Therapeutic Design	Epitope-focused vaccines (VLP, subunit, mRNA); bispecific antibody engineering; iterative preclinical testing	LLM generative design of immunogens/ antibodies; agent docking and MD simulations; AI refines candidates iteratively with minimal human intervention

Generated by local LLM (gpt-oss:20b)

Activity Topic	Traditional Workflow	Proposed AI-Enabled Workflow
1. Literature review & data curation	- Manual search of PubMed, PMC, pre-print servers, and relevant journals (e.g., <i>Cell Reports</i>, <i>Nature Communications</i>). - Extraction of sequence, structure, and epitope data by hand; entry into spreadsheets or custom databases. - Cross-checking with databases (GenBank, UniProt, PDB, IEDB, ViPR).	- Retrieval-Augmented Generation (RAG) system populated with the same sources but indexed for instant query. - LLM (e.g., GPT-4-based) receives natural-language prompts ("List all known broadly neutralising antibodies against CHIKV E2 domainB") and returns structured JSON (antibody name, PDB ID, epitope residues). - Agentic scripts automatically pull new entries from APIs, update the RAG corpus, and flag gaps for the researcher.
2. Sequence & structure acquisition	- Download FASTA and PDB files manually or via batch scripts; store locally. - Manual bookkeeping of accession numbers and metadata.	- Agent "Sequence Fetcher" commanded by the LLM to query GenBank and UniProt for all alphavirus E2 sequences, automatically normalising gene names. - Agent "Structure Collector" pulls all relevant PDB/EMDB entries (e.g., 6I0P, 7C3P) and archives them in a project-specific repository.
3. Multiple sequence alignment & phylogeny	- Run MAFFT or ClustalQ on the command line, then visually inspect with Jalview. - Generate phylogenetic trees in MEGA or RAxML, manually interpreting clades.	- LLM translates a goal ("Identify surface-exposed residues conserved across Old-World alphaviruses") into a set of commands for an Alignment Agent and a Phylogeny Agent. - The agents produce an MSA and a maximum-likelihood tree, returning metadata (conservation scores, bootstrap values) in machine-readable format.
4. Structural mapping of conserved residues	- Load PDB in PyMOL/ChimeraX; use scripts to compute solvent-accessible surface area (SASA) and residue conservation overlays. - Manually correlate SASA with sequence conservation to propose candidate epitopes.	- Agent "Structure Annotator" automatically maps conservation scores onto the 3-D structure, generates a per-residue heat map, and flags surface patches above a conservation threshold. - LLM interprets the map and produces a short report highlighting likely epitope clusters.
5. Epitope extraction & mapping	- Extract experimental epitope data from literature (e.g., peptide-array results, HDX-MS, cryo-EM), manually transcribe into tables. - Cross-reference with IEDB entries and align to structures.	- RAG-LLM pulls all IEDB, PDB, and literature-derived epitope annotations for the selected viruses. - LLM returns a unified epitope catalogue (sequence, structural coordinates, antibody partner, neutralisation potency) and flags inconsistencies for curation.
6. Paratope extraction & characterization	- Manually parse antibody heavy/light chain sequences, CDR loops, and structural data (e.g., from PDB). - Annotate V-gene usage, CDR length, key contact residues.	- LLM queries the RAG corpus for all bNAb structures against alphaviruses, extracts CDR sequences and 3-D paratope footprints, and compiles a machine-readable dataset. - Agent "Paratope Analyzer" computes contact maps, hydrogen-bond networks, and electrostatic complementarity automatically.
7. Consensus paratope identification	- Perform clustering of paratope sequences (e.g., via CD-HIT) and manually inspect clusters for common motifs. - Use statistical tests to link motif presence with breadth of neutralisation.	- LLM formulates a supervised ML task (classification of breadth) and hands the data to a graph-neural-network agent. - Agent trains a model, outputs feature importance (e.g., V-gene usage, specific CDR residues) and proposes a "consensus paratope" motif.
8. In silico design of immunogens/antibody variants	- Engineer epitope scaffolds or V-gene grafts by hand in Rosetta or PyRosetta, then run energy minimisation. - Validate predicted binding by running docking (HADDOCK, AutoDock) and inspecting results manually.	- LLM acts as a generative design engine: it proposes chimeric immunogens or CDR grafts that retain the consensus paratope features. - Agent "Docking-Sim" automatically performs rigid-body docking, then MD simulations (e.g., GROMACS) to assess stability, feeding results back to the LLM for iterative refinement.
9. Experimental validation pipeline	- Design ELISA/neutralisation assays, order peptides/proteins, and conduct assays in the lab. - Record results in a spreadsheet, perform statistical analysis with GraphPad Prism.	- LLM drafts experimental protocols (e.g., peptide array layout, pseudovirus neutralisation assay) and produces QC checklists. - Agent "Lab Scheduler" interfaces with automated liquid handlers and high-throughput sequencers, uploading raw data to a LIMS that is indexed by the RAG system for downstream analysis.
Key terminology		
Term	Definition	
LLM	Large Language Model (e.g., GPT-4), a transformer-based neural network trained on vast corpora of text.	
RAG	Retrieval-Augmented Generation; a framework that augments a language model's generative capacity with a retrieval engine that fetches relevant documents in real time.	
Agentic workflow	A paradigm in which an LLM issues high-level directives to specialized software "agents" (e.g., alignment, docking, data-fetching scripts) that execute concrete computational tasks.	
Consensus paratope	A set of structural and sequence features (e.g., V-gene, CDR length, anchor residues) common to multiple broadly neutralising antibodies targeting the same epitope.	
Epitope	The part of an antigen that is recognised by an antibody; can be linear (continuous) or conformational (discontinuous).	

Hackathon Q&A To Address (Day 1)

1. What is our research goal / question?

Our goal is to explore how large language models (LLMs), particularly locally-deployed ones, can serve as **AI research assistants** to support scientific discovery—specifically in the context of **alphavirus vaccine research**.

We aim to test tools like Retrieval-Augmented Generation (RAG) and AI Agents to see if they can help researchers query literature, organize insights, and potentially accelerate biomedical research workflows.

2. What data will we investigate? What criteria do we use for choosing it?

We focus on two research papers about alphaviruses, particularly **chikungunya virus (CHIKV)** and its **E2 glycoprotein**—a critical surface protein targeted by neutralizing antibodies.

These documents were chosen because:

- They are **scientifically relevant** to an ongoing global health challenge.
- They **include structural biology and immunological data**, ideal for testing LLMs' ability to process and summarize complex biomedical information.
- The content is technical and domain-specific, making it a good test case for evaluating the usefulness of AI assistants.

3. Tools - reagents, software, websites

We use the following tools:

- **LLMs:** ChatGPT (Agent Mode), LLaMA 3.2 (local)
- **Local LLM platforms:** Ollama (CLI), LM Studio (GUI), GPT4All
- **Techniques:** Retrieval-Augmented Generation (RAG), local agent frameworks
- **Reference materials:** 2 PDF scientific papers on alphaviruses (hosted on shared Google Drive)

4. What instructions do we need?

We guide the LLMs with structured prompts such as:

- "Summarize the main findings about the E2 glycoprotein."
- "List known neutralizing antibody targets."
- "Compare the vaccine development strategies between paper A and paper B."

We also design custom prompt templates to help evaluate information extraction, comparison, and hypothesis generation.

5. What needs to be tracked and recorded during the project?

- **Responses** of different LLMs (ChatGPT vs local models)
- **Prompt effectiveness** (how clearly the AI followed instructions)
- **Accuracy and depth** of scientific insights retrieved
- **Ease of deployment** and **performance** of local LLM tools
- **Team reflections** on usability and workflow impact

6. What questions need to be answered?

- Can LLMs assist with parsing complex biomedical research?
- Are local LLMs viable tools in educational or low-resource settings?
- How effective are RAG and agent-based frameworks in scientific workflows?
- What are the limitations in domain specificity, reasoning, or fact extraction?

7. How will the work be assessed?

- Qualitative and/or quantitative assessment of insight quality and completeness from AI outputs
- Comparison across tools (ChatGPT vs LLaMA 3.2 via Ollama or LM Studio)
- Usability and reproducibility by team members with different tech backgrounds
- Feedback from team consultants and subject matter experts on scientific accuracy

